(Agentic) Software Engineering
BU CAS CS 413 Fall 2026

(Syllabus)

- **Semester** Fall, 2026
- **Teaching Staff**
  - Instructor – Hongwei Xi
  - Teaching Assitant – Victor Wang
    - Office Hours: TBA
    - Location: CDS 1013, e-mail: `hwxi@cs.bu.edu`
- **Lecture Times**: TR 12:30-1:45PM
- **Classroom**: CAS 211
- **Reference Books**:
  - Software Engineering by Ian Sommerville, **9th** edition, Pearson/Addison-Wesley, 2011.
    - (ISBN: 978-0-13-703515-1)
    - *The textbook provides the conceptual foundation for the course. Some topics involving current development practices will be supplemented with additional readings and examples.*
- **Homepage**: `https://hwxi.github.io/TEACHING/CS413/2026Fall`
- **Description**: This course introduces the principles and practices used in the engineering of medium/large-scale software systems. It focuses on software development as a disciplined and collaborative engineering activity rather than simply the construction of programs.

  Primary topics include software processes, agile development, requirements engineering, system modeling, architectural and object-oriented design, software testing, software evolution, dependability and security, software reuse, project management, quality management, and configuration management.

  A major component of the course is a semester-long team software project (involving the construction of a compiler for a small functional programming language). Students apply the techniques discussed in class to the specification, design, implementation, testing, and evolution of a nontrivial software system.

  The course uses Ian Sommerville's Software Engineering, 9th edition, as its primary text while supplementing the book with selected contemporary software engineering practices.
- **Prerequisites**: This course is designed for students who already have an immediate level of proficiency in Python. You are expected to be familiar with a text editor (e.g., vim, emacs, vscode).

- **Exams**:
  - First midterm exam: TBA
  - Second midterm exam: TBA
  - The final exam date is yet to be anounced.
- **Grades** The final score is calculated using the following formula.

  $$\text{final score} = 30\%\cdot(\text{homework}) + 20\%\cdot(\text{midterm}) + 40\%\cdot(\text{final}) + 10\%\cdot(\text{participation})$$

  The final letter grade is calculated as follows.
  - **A**: final score is 90% or above
  - **B**: final score is 80% or above
  - **C**: final score is 70% or above
  - **D**: final score is 60% or above
- **Program Submission**: Homework assignments are to be submitted via the Gradescope system.
- **Attendance**: It is expected that you will attend the lectures.
- **Academic Integrity**: We adhere strictly to the standard BU guidelines for academic integrity. For this course, it is perfectly acceptable for you to discuss the general concepts and principles behind an assignment with other students. However, it is not proper, without prior authorization of the instructor, to arrive at collective solutions. In such a case, each student is expected to develop, write up and hand in an individual solution and, in doing so, gain a sufficient understanding of the problem so as to be able to explain it adequately to the instructor. Under *no* circumstances should a student copy, partly or wholly, the completed solution of another student. If one makes substantial use of certain code that is not written by oneself, then the person must explicitly mention the source of the involved code.